STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I. for Gas Service	**Case 19-G-0310**

Prepared Direct Testimony of

New York City Gas Infrastructure and Safety Panel

Michael McCutcheon
Ke Wei
Benjamin Krakauer
Pinar Balci, Ph.D.
Chief David Morkal

On Behalf of:

The City of New York

August 30, 2019

INTRODUCTION

Q. PLEASE STATE YOUR NAME, OCCUPATION AND BUSINESS ADDRESS.

A. Mr. McCutcheon – My name is Michael McCutcheon. I am currently a Pipeline Coating consultant for Quality Integrated Services, Inc. My business address is 35 North Church Street, Cortland, New York 13045.

Ms. Wei – My name is Ke Wei. I am Assistant Director for Infrastructure for the New York City Mayor's Office of Sustainability and New York City Mayor's Office of Resiliency. My business address is 253 Broadway, 14th floor, New York, New York.

Mr. Krakauer – My name is Benjamin Krakauer. I am the Assistant Commissioner for Strategy & Program Development at New York City Emergency Management, or NYCEM. My business address is 165 Cadman Plaza East, Brooklyn, New York.

Ms. Balci – My name is Pinar Balci. I am the Assistant Commissioner in the Bureau of Environmental Planning and Analysis at the NYC Department of Environmental Protection or DEP. My business address is 59-17 Junction Boulevard, Floor 11, Flushing New York.

Chief Morkal – My name is Battalion Chief David Morkal. I am the Executive Officer to the Chief of Operations for the Fire Department of New York, or FDNY. My business address is 9 Metrotech Center, Brooklyn, New York 11201.

Q. ON WHOSE BEHALF ARE YOU SUBMITTING THIS TESTIMONY?

A. We are submitting this direct testimony before the New York Public Service Commission ("PSC" or the "Commission") on behalf of the City of New York ("City").

Q. PLEASE DESCRIBE YOUR EDUCATIONAL BACKGROUND.

A. Mr. McCutcheon – I hold a number of certifications from a variety of professional organizations, including the American Petroleum Institute (Certified Pipeline Inspector) and the American Welding Society (Certified Welding Inspector). I have completed all training necessary to perform audits for Pipeline and Hazardous Materials Safety Administration. In 2004, I completed a certification from the New York State Fire Academy for fire behavior, arson investigation, and fire investigation. I have also completed the NACE International Coating Inspector Program, Level 1.

Ms. Wei – I received a Bachelor of Arts degree in International Relations from Pomona College and a Master in International Affairs with a focus in Energy Management and Policy from Columbia University School of International & Public Affairs.

Mr. Krakauer – I hold a Master's Degree in Public Administration from Binghamton University's College of Community and Public Affairs and a Baccalaureate Degree in Political Science from Binghamton University's Harpur College of Arts and Sciences. Additionally, I am a graduate of the Corning

Community College/Arnot-Ogden Medical Center Emergency Medical Technician-Paramedic Program.

Ms. Balci – I received a bachelor of science from Hacettepe University in Turkey in 1994 and I received a master of science from Hacettepe University in Turkey in 1996. Subsequently, I received a doctorate of philosophy from the University of North Texas in 2002. Following, I received a post-doctorate from Murray State University in Kentucky in 2004.

Chief Morkal – I received a Bachelor of Fine Arts from the University of Evansville in 1981. I am a graduate of the FDNY's Fire Officers Management Institute, Advanced Leadership Course, West Point Combating Terrorism Leadership Program, L-380 "The Point of the Spear", L-381 "Incident Leadership" L-481, and L-580, Staff Ride at Gettysburg National Military Park.

Q. PLEASE DESCRIBE YOUR PROFESSIONAL EXPERIENCE AND CURRENT RESPONSIBILITIES.

A. Mr. McCutcheon – I have over 42 years of practical field experience and knowledge construction of liquid and natural gas systems, with experience in all phases of pipeline construction, from large transmission to smaller distribution systems, from wellhead to meter. From 1985 to 2015, I was employed by the NYS Department of Public Service as a Utility Analyst for both regulatory compliance and incident investigation. In this role, I was the lead investigator for 37 incidents, conducting a code and document review for culpability, cause and origin, facility testing, evidence procurement, and root cause analysis. I also conducted inspections for

safety and regulatory compliance of pipeline facilities within New York State, which included manual review, field observation and trailing audits, construction monitoring and review. Since 2016, I have served as an inspector and quality assurance/quality control inspector for several companies, including in my current position as a Pipeline Coating consultant for Quality Integrated Services, Inc.

Ms. Wei – I have been in my current position since January 2016. In this role, I advance and manage regulatory and policy issues and relevant research related to sustainability and resiliency of energy supply and distribution across electricity, natural gas, and fuels. I have spoken at various local, regional, and international conferences and workshops on energy resiliency and sustainability issues. Prior to joining the City, I was a Program Director at Columbia University's Center on Global Energy Policy, including managing their fuel resilience research program. Before that, I was a management consultant with McKinsey & Company.

Mr. Krakauer – My career in emergency services began more than 20 years ago as a volunteer for a local fire department and rescue squad in Suffolk County, New York. Through high school and college, I worked as a Certified First Responder, Emergency Medical Technician, Advanced Emergency Medical Technician – Critical Care, Paramedic, and emergency medical services instructor for a number of departments throughout New York State, including serving as the Chief/Executive Director of a volunteer ambulance service and serving as a Manager and Controlled Substances Officer for a commercial ambulance service that provided coverage to two counties in upstate New York. In 2007, I was

appointed as the Emergency Preparedness Coordinator for Broome County, New York where I focused on all-hazards emergency planning, emergency operations center operations, and public health preparedness. I served as the Incident Commander during Broome County's response to the H1N1 Influenza Pandemic and served as the Planning Section Chief and Deputy Emergency Manager during the American Civic Association mass shooting. In 2010, I joined NYCEM as a Public Warning Specialist, working in Watch Command, sending out emergency messages to the public, and building the City's emergency public notification program, a role which I held for almost two years. I then spent the next six years as the Deputy Director and then Director of NYCEM's Watch Command, where I was responsible for the strategic and operational direction of NYCEM's 24x7 operations center, on-duty emergency managers, and the Notify NYC program. I am currently the Assistant Commissioner for Strategy and Program Development where I oversee intra- and interagency high priority initiatives on behalf of the Commissioner and First Deputy Commissioner, including the citywide mass notification contract, the on-call emergency contracts program, and the interim flood protection measures program. I serve as the leader of one of NYCEM's three on-call teams, consisting of 50 professionals responsible for coordinating the City's response to emergencies and disasters and leading the Emergency Operations Center, or EOC, when activated.

Ms. Balci – I have been with DEP for over seven years. I oversee and manage multiple programs, such as the Green Infrastructure Program, the Municipal

Separate Storm Sewer System Program, integrate planning, and provide City Environmental Quality Review reviews for water and wastewater infrastructure projects. Prior to joining DEP, I worked at the South Florida Water Management District, where I managed the Federal and State Policy and Permitting Division to achieve the goals of the Everglades restoration and capital projects. I also worked as Northern Everglades Implementation Manager overseeing and coordinating all planning, permitting and capital project activities.

I specialize in management, coordination and implementation of water resources and environmental restoration projects. I am familiar with the range of pertinent environmental regulations including the National Environmental Policy Act, the Endangered Species Act, the Clean Water Act and associated requirements.

Chief Morkal – I am a 29-year veteran with the FDNY. I was promoted to Lieutenant the week after the World Trade Center attacks, where I worked in Ladder 18 on the Lower East Side. After being promoted to Captain I was transferred to Engine 23 in Midtown Manhattan. As a Battalion Chief I was assigned to Battalion 4 and worked as the Chief in Charge of Probationary Firefighter School for 2 years. I also worked as the Planning Section Chief for the FDNY Incident Management Team and I am currently one of the Deputy Incident Commanders. I have been activated with the team for Hurricane Gustav in Louisiana, Hurricane Irene in both New York City and Binghamton, New York, Hurricane Sandy in New York City, the snow storm in Buffalo, New York, the East Harlem explosion and collapse, and several wild land fires in the Southwest

with other Incident Management Teams. I have served in my current position as Executive Officer to the Chief of Operations since March 2019.

Q. DO YOU HAVE PRIOR EXPERIENCE TESTIFYING?

A. Mr. McCutcheon – Over the last 30 years, I have made public presentations and been on panels for public information mostly related to damage prevention. I have not previously testified before the PSC in a utility rate case.

Ms. Wei – I speak frequently before international, national and local audiences on the City's energy policy and resilience goals. I also testified in the ongoing Consolidated Edison Company of New York, Inc. electric rate case, Case 19-E-0065, as a member of the New York City Electric Infrastructure Panel.

Mr. Krakauer – I have testified before the New York City Council and a U.S. House of Representatives subcommittee on providing emergency warnings to the public. I also testified on behalf of the City in the ongoing Consolidated Edison Company of New York, Inc. electric rate case, Case 19-E-0065.

Ms. Balci – I do not have prior experience testifying.

Chief Morkal – I have testified before the New York City Council on behalf of the FDNY. I have not previously submitted testimony to the PSC.

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. Our testimony will cover five broad areas pertaining to National Grid's gas rate filing. We comment on The Brooklyn Union Gas Company d/b/a National Grid NY's ("KEDNY") and KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I.'s ("KEDLI"; and together with KEDNY, "National Grid" or the "Companies") Gas

Infrastructure and Operations Panel's ("GIOP") pre-filed direct testimonies and their associated exhibits, supplements, and updates. We also comment on National Grid's Gas Safety Panel's pre-filed direct testimony and its associated exhibits, supplements, and updates. Our testimony focuses on the following issues and recommendations:

- National Grid is proposing significant increases in capital spending, nearly tripling from its 2014 spending, with even larger increases in specific programs, namely Leak Prone Pipe replacement and City/State Construction ("CSC"). In light of the increased spending, the programs within the capital budget must be reviewed critically. To protect ratepayers from these significant increases (if the increases are justified), the City recommends two proposals. First, the Commission should continue to require a reconciliation mechanism so that any amounts unspent are returned to customers. Second, the Commission should reject the Companies' proposed changes to the CSC tracker and O&M reconciliation mechanisms.
- National Grid is proposing several storm hardening and system safety projects, including upgrades to the Greenpoint LNG Plant and the proposed installation of remote service shutoff valves. While the City is conceptually supportive of these proposals, they require additional study, greater quality control, or more coordination with the City to improve their effectiveness. Beyond what National Grid has proposed, the City also proposes that

National Grid implement recommendations to harden priority mini-gate and take stations that were contained in the Companies' Storm Hardening Collaborative Report.

- The City is making efforts to offset the deleterious effects of Combined Sewer Overflows (CSOs) through Green Infrastructure Assets, in order to comply with federal clean water act requirements that are enforced by the State. These assets have suffered damage as a result of National Grid actions/inactions. Therefore, the City asks that National Grid coordinate with the City and receive the appropriate training to prevent further damage to these Assets.
- KEDNY and KEDLI propose numerous changes to their array of gas safety-related metrics, including changes to both positive and negative revenue adjustment mechanisms. The City opposes many of these changes as they either seek to reward shareholders for business-as-usual performance or merely reduce the likelihood that the Companies will be subject to negative revenue adjustment without material changes in behavior. Accordingly, the City recommends a series of changes to the Companies' proposed incentive mechanisms to encourage more desirable outcomes.
- Finally, both Companies have a substantial balance of regulatory liabilities. In lieu of using these liabilities to pay for other specific programs, as suggested by National Grid, the full balance of these accounts, with the appropriate carrying charges, should instead be returned to ratepayers.

Q. ARE YOU SPONSORING ANY EXHIBITS WITH YOUR TESTIMONY?

A. Yes. We offer Exhibit__(NYCGISP-1), which is a compilation of the discovery responses from National Grid that we relied upon in preparing our testimony, some of which we expressly reference herein.

Exhibit__(NYCGISP-2), which is a compilation of relevant regulations and practice guidelines that we relied upon in preparing our testimony, some of which we expressly reference herein.

Exhibit__(NYCGISP-3), which is a summary of our proposed incentive mechanisms and safety metrics.

CONCERNS WITH THE OVERALL CAPITAL PROGRAM

Q. DO YOU HAVE ANY CONCERNS WITH NATIONAL GRID'S COMPREHENSIVE SUITE OF PROPOSED CAPITAL PROJECTS?

A. According to each Company's GIOP pre-filed testimony, the volume of the Companies' planned capital work continues to increase significantly from 2020 through 2024. The capital work is driven primarily by an aggressive Proactive Main Replacement ("LPP") Program, and KEDNY's CSC Program. The major drivers for the increase in gas capital expenditures in the Rate Year (which are also generally applicable to Data Years ("DY") 1, 2, and 3) include: the LPP Program ($485.3 million), CSC Program ($297.1 million), specific infrastructure upgrades, including KEDNY's Metropolitan Reliability Infrastructure Project and KEDLI's Northwest Nassau Project ($199.2 million, total), and System Integrity Programs ($19.0 million). KEDNY sets forth its major programs on pages 25-53 of its GIOP

testimony, and KEDLI addresses its major programs on pages 25-44 of its GIOP testimony. Both KEDNY's and KEDLI's Exhibit___(GIOP-1CU) detail these major drivers, as well.

Q. WHAT ARE YOUR SPECIFIC CONCERNS WITH THESE PROPOSED INVESTMENTS?

A. We are concerned that this level of capital expenditure is not realistic, and that National Grid may not be able to prudently increase its spending by such amounts while maintaining effective and efficient management, supervision, and cost controls to obtain the desired result of the various programs and projects. We also are concerned as to whether National Grid – as well as its external contractors – has the adequate skilled personnel to competently and safely perform all of the proposed capital work in accordance with all applicable regulations.

Q. WHAT IS THE BASIS FOR YOUR CONCERNS?

A. As mentioned above, the proposed expenditures continue to increase dramatically. The Companies' combined past actual capital spending inclusive of cost of removal, as seen in the response to DPS-84, was $488.7 million in 2014, $600.7 million in 2015, $729.0 million in 2016, $769.7 million in 2017, and $1.04 billion in 2018. In the current proceeding, National Grid is outlining a capital budget proposal of $1.42 billion in the rate year. This is nearly triple the capital spend from 2014.

As the level of spend increases and these programs get larger, it becomes more difficult to manage, requiring more resources, materials and coordination

effort. This places strain on both internal and external resources. Additionally, these programs require new processes and procedures to be effectively managed and remain sustainable for the long term. Thus, it is important that appropriate project controls are in place to ensure prudent use of capital and that programs are managed effectively. We acknowledge that National Grid is proposing to fill approximately 252 and 495 incremental positions by the end of the Rate Year and Data Years, respectively. The Commission should ensure that the Companies have adequate project management staff to handle the proposed increase in capital expenditures effectively.

Q. BEYOND THE MAGNITUDE OF THE COST INCREASES, DO YOU HAVE ANY OTHER CONCERNS?

A. Yes. Recently, the Commission issued its Order Instituting Proceeding and to Show Cause (the "Order to Show Cause") in Case 17-G-0317. The allegations have not been proven; however, they are credible enough for the Commission to require National Grid to answer for them. The sheer number of alleged violations – 1,616 so far, with possibly more to come, according to page 14, fn. 22 of the Order to Show Cause – and the nature of the alleged violations raises legitimate concerns over whether National Grid can safely and prudently manage a capital budget of this size.

Q. WHAT ABOUT THE NATURE OF THE ALLEGED VIOLATIONS RAISES CONCERNS ABOUT PRUDENT SYSTEM MANAGEMENT?

A. These alleged violations, if proven true, are fundamentally safety violations resulting from a material oversight failure by National Grid. Pages 13-14 of the Order to Show Cause concisely summarize the nature of the alleged violations. The New York State Department of Public Service Staff found: (i) that the Queens Pipeline had been installed improperly, "either in very close proximity to, or directly in contact with, various other underground utilities;" (ii) installation of mains and main replacements were done by unqualified contractors; and (iii) joining work was completed by unqualified workers. That these alleged violations arose from unqualified contractors and not National Grid employees does not exculpate the Companies. The consequences of a gas pipeline rupture are the same regardless of whether a contractor or employee violated safety regulations. Moreover, National Grid is responsible for managing its contractors as well as its systems. If National Grid wants to increase the scale and scope of its capital programs at ratepayer expense, it must first demonstrate that it can do so safely and prudently, including identifying gaps in its current project management and oversight approach, and developing strategies to remediate those gaps to ensure that safety concerns are avoided or mitigated.

Q. WHAT ARE YOUR RECOMMENDATIONS?

A. In our view, the Main Replacement and Service Replacement Programs should be planned to more thoroughly take advantage of bundling projects, minimizing disruptions to the communities, and ensuring construction is as efficient as possible, thereby reducing the overall program costs. We recommend that the

Companies' risk model for main replacement be reviewed annually to ensure maximum optimization of main work and that the Companies continue to identify opportunities to coordinate with City projects involving street works.

Additionally, the Companies must be able to exhibit that the safety and reliability of their systems is improved at a rate commensurate with the proposed spending, and that the capital programs are prudently managed at a level commensurate with the spending. If that cannot be shown, then the main replacement programs should be decelerated to an appropriate rate. Finally, given the magnitude of the capital budgets the Companies are requesting, we recommend that the Commission require a reconciliation mechanism so that any amounts unspent are returned to customers.

Q. ARE THE COMPANIES PROPOSING ANY CHANGES TO EXISTING RECONCILIATION MECHANISMS?

A. Yes, they are. The Companies propose to continue the downward-only net plant capital tracker that was approved in the 2016 KEDNY and KEDLI Rate Cases. We support this proposal. However, the Companies also have an existing CSC tracker mechanism that currently requires a 90 percent deferral of the difference between forecast and actual CSC spending. The Companies propose to modify this tracking mechanism to a full reconciliation of the difference between forecast and actual spending. KEDNY also proposes to add an O&M tracker "that would fully reconcile the difference between forecast and actual spending in the non-

reimbursable "Support and Protect" O&M expense category" which are certain non-reimbursable CSC O&M expenditures.

Q. DO YOU AGREE WITH THE PROPOSED CHANGES TO THE CSC-RELATED RECONCILIATION MECHANISMS?

A. No. The Companies state that the proposed CSC tracker changes and the new Support and Protect O&M expense tracker is based on the "increasing volatility and scope of unplanned CSC projects…" However, the City's Policy Panel testimony, which is being filed concurrently with this testimony in these cases, is recommending that National Grid designate a high-ranking official to serve as a designated liaison with the City specifically with respect to CSC projects, and that National Grid submit regular reports to the Commission regarding the status of pending CSC projects so that performance can be objectively tracked over time. These recommendations, if adopted, should help address the Companies' stated concerns over increased volatility and scope surrounding CSC. Therefore, the proposed changes to the CSC tracker are not necessary.

In addition, given the size and scope of the CSC program, a full reconciliation mechanism introduces additional, unknown ratepayer exposure. For example, KEDNY has a CSC budget of almost $300 million per year, so even a 10 percent variation between forecasted and actual costs could have significant ratepayer impacts. The Companies should not be insulated from these variations, and doing so would reduce the Companies' incentives for accurate forecasts and effective cost management. Indeed, the Companies themselves recognize the

importance that accurate forecasting plays within the ratesetting process, particularly as it relates CSC costs. In response to DPS-352, KEDNY argues for setting rates using its forecast of actual CSC costs, noting that the "CSC tracker is…not a substitute for setting rates." The Companies' proposal for a full CSC reconciliation mechanism would, however, create just such the substitute that the Companies claim they want to avoid.

For these reasons, the Commission should reject the proposed changes to the Companies' CSC tracker and reject the similar, but related, O&M tracking mechanism.

CONCERNS WITH SPECIFIC CAPITAL PROGRAMS

A. LEAK PRONE PIPE

Q. DO YOU HAVE ANY CONCERNS REGARDING THE COMPANIES' LPP REPLACEMENT PLANS?

A. Yes. The Companies request rate allowances to replace the following miles of LPP in calendar years 2020-2023:

- KEDNY: 70, 75, 80, 85;
- KEDLI: 155, 155, 155, 155.

Minimum removal targets for KEDNY to avoid negative revenue adjustments ("NRAs") are proposed to continue to increase by 5 miles per year. For 2019-2022, minimum targets would be: 60, 65, 70, and 75, with a cumulative minimum of 310 miles for calendar years 2020-2023. Minimum removal targets for KEDLI are held steady at 150 miles per year. The Companies further propose:

- Continuing an existing incentive of two basis points ("BPs") for each full mile of LPP removed above the rate allowance thresholds, capped at 10 BPs.
- Continuing an existing surcharge mechanism (the Gas Safety and Reliability Surcharge, or "GSRS") that allows the Companies to recover costs of retired LPP incremental to the level funded in base rates.

Q. WHAT CONCERNS DO YOU HAVE WITH THESE PROPOSALS?

A. National Grid plans to spend an average of $571.7 million annually on its LPP Program, with an additional $19.7 million annual average on its Service Replacement Program, from 2020 to 2024 (comprising the rate year and data years). These programs are for the replacement of leak prone gas mains and services (defined as small diameter (12" and smaller)) that are: (i) cast and wrought iron; (ii) unprotected steel (non-cathodically protected; (iii) unprotected steel/wrought iron services; and (iv) copper and pre-1974 high-density polyethylene services.

We evaluated the programs and have identified that the combined KEDNY and KEDLI costs associated with these programs have significantly increased from 2014 to the proposed Rate Year spending levels by approximately 645% for the LPP program. The majority of the increases are associated with increased mileage of pipe and services replaced, asphalt and paving of roads and curbs due to the replacement, as well as higher overheads.

Q. PLEASE CONTINUE.

A. As noted above, the Companies' budgets for the LPP Program have reached historic levels, and those levels will only continue to increase under the Companies' proposals. While we agree with the Companies that it is important to continue removing LPP, as doing so is consistent with Commission policy and will reduce leak rates on the Companies' systems, LPP removal efforts must be balanced against customer impacts.

As a result, we support the Companies' requested rate allowances for proactive LPP removal activities, but the requests for an incentive for additional miles above the rate allowance threshold and the request to fund additional miles replaced via the GSRS should be rejected. To the extent that either Company incurs incremental costs to replace additional LPP miles above and beyond the rate allowance threshold, the costs associated with those replacements should be treated similar to other capital projects and deferred for examination and possible recovery in a future rate case.

Moreover, because the Companies' LPP Programs are now designed to replace or retire all LPP within the 20-year horizon called for under Commission policy, there is no need to provide shareholder incentives to the Companies to the extent that either Company exceeds the rate allowance threshold; the Companies do not need an incentive to conform to Commission directives. The existing incentive for each full mile of LPP removed above the rate allowance thresholds should therefore be discontinued.

Q. SHOULD THE COMMISSION CONTINUE THE EXISTING POSITIVE REVENUE ADJUSTMENT IF THE COMPANIES MEET SPECIFIC UNIT COST TARGETS RELATED TO LPP REMOVALS?

A. No. The Companies currently have an incentive mechanism that allows the Companies to earn a positive revenue adjustment ("PRA") of up to 10 BPs if the Companies can meet specific unit cost targets associated with LPP replacements. The Companies propose to continue this incentive mechanism as it currently exists, which means that the unit cost targets embedded in the incentive mechanism would be updated to reflect the unit costs for LPP replacements that are embedded in rates. KEDNY proposes that it start earning incentives for unit costs at or below $1,285/foot, and KEDLI proposes that it start earning incentives for unit costs at or below $306/foot. As a point of comparison, in the current Rate Plan for CY 19, which was set in 2016, KEDNY's incentives begin at $664/foot, and KEDLI's begin at $238/foot. During the current rate plan, the unit costs at which the Companies would begin to earn incentives only increases year-over-year by between 2.5% and 5%.

Q. WHAT ARE YOUR SPECIFIC CONCERNS WITH THIS PROPOSAL?

A. The Companies claim that unit costs for LPP replacements have increased significantly since the last rate case. Thus, for purposes of the LPP unit cost PRA, the Companies argue that the unit cost targets that serve as the starting point for determining whether the Companies have achieved any unit cost savings should also increase under the Companies' proposal.

Under the Companies' proposal, the point at which the Companies would begin earning incentives would be updated to $1,285/foot for KEDNY and $306/foot for KEDLI, respectively, which between CY19 and FY21 represent a 93.5% increase in KEDNY's unit costs and a 28.6% increase in KEDLI's unit costs. While we do not comment on the specific drivers behind the alleged unit cost increases, the proposed threshold resets under the Companies' proposed PRAs nevertheless represent dramatic increases in starting points for both Companies, which raises fundamental questions about the appropriateness of continuing this mechanism 'as-is.' As demonstrated in the charts below, KEDNY in particular has struggled to come anywhere close to earning the productivity incentive under the current rate plan, with actual unit costs in 2017 and 2018 nearly doubling the minimum unit cost incentive thresholds. KEDLI's performance has been better but it, too, has been unable as of yet to achieve any incentives under this mechanism.

KEDNY Historical LPP Removal Productivity Performance (DPS-494)

Year	Minimum incentive threshold (2 BP)	Highest incentive threshold (10 BP)	Actual unit cost (approximate)	Incentive earned (BP)
2017	$618/ft	$580/ft	$630/ft	No (0)
2018	$648/ft	$609/ft	$846/ft	No (0)

KEDLI Historical LPP Removal Productivity Performance (DPS-597)

Year	Minimum incentive threshold (2 BP)	Highest incentive threshold (10 BP)	Actual unit cost (approximate)	Incentive earned (BP)
2017	$223/ft	$209/ft	$250/ft	No (0)
2018	$232/ft	$218/ft	$272/ft	No (0)

Q. IS INCREASING THE THRESHOLD AT WHICH THE COMPANIES MAY EARN AN INCENTIVE IN RATEPAYERS' INTERESTS?

A. No. We certainly support cost-effectiveness, as this PRA seeks to incentivize. However, the Companies' modifications do not incentivize cost-effectiveness. Rather, the modifications requested by the Companies essentially seek to reset the target to move the starting point closer to the Companies' actual per-foot costs to make this incentive easier to achieve. The Companies thus propose to modify this measure to reward business-as-usual behavior instead of rewarding meaningful main replacement cost reductions. Ratepayers, under this proposal, would have to pay for unit costs only a modicum less expensive than the current unit costs and reward the Companies for doing so. Narrowly beating current actual costs that far exceed prior forecasts should not require incentives.

Q. WHAT DO YOU RECOMMEND?

A. The threshold at which the Companies may receive an incentive should remain the same as it is in the current Rate Plan and increase yearly at no more than the same rate than which it increased under the current Rate Plan (e.g., between 2.5-5% per year). As shown in Exhibit___(NYCGISP-3), we are proposing a 3.7% yearly unit cost increase for KEDNY and a 3.5% yearly unit cost increase for KEDLI, which is consistent with the average yearly increases embedded within the current Rate Plan. This adequately rewards cost-effective solutions while ensuring that ratepayers do not reward the Companies for business-as-usual performance. If the Companies contend that an incremental increase to the current unit cost threshold renders the incentive meaningless because it is too difficult to achieve, then this incentive mechanism should be discontinued entirely. LPP unit costs have been rising significantly. The table below, which compiles the responses to DPS-494 and DPS-597, illustrates these increases.

KEDNY and KEDLI LPP Unit Costs ($/ft)

	FY 2015	**FY 2016**	**FY 2017**	**FY 2018**	**FY 2019**
KEDNY	614	478	630	846	1031
KEDLI	227	207	250	272	327

Alternatively, given the significant increases in LPP unit costs in recent years, the prudent course may be to suspend this incentive mechanism for the time being and revisit it once replacement costs have stabilized.

Q. DO YOU HAVE ANY CONCERNS REGARDING THE COMPANIES' REPLACEMENT OF LEAK-PRONE PIPE IN FLOOD-PRONE AREAS?

A. Yes.

Q. WHY IS THE COMPANIES' REPLACEMENT OF LEAK-PRONE MAINS AND SERVICES IN FLOOD PRONE AREAS IMPORTANT?

A. During a flood, the pressure of the flood waters outside of the gas main can be greater than the pressure in the gas main, especially as it pertains to low pressure mains and services. This causes water to enter through any of the leaks (*e.g.*, cracks, holes) present in the pipe. Once there is water intrusion in the mains, it can be a lengthy process to dewater the main, which can require shutting off gas to the affected main and relighting the system.

Q. HOW MUCH LPP DOES EACH COMPANY CURRENTLY HAVE WITHIN THE CURRENT 2015 FEMA FLOODPLAIN?

A. According to the responses to CNY-46 and CNY-47, as of December 31, 2018, KEDNY and KEDLI had the following miles of LPP located within the 2015 FEMA floodplain:

	KEDNY (miles)	**KEDLI (miles)**
Low Pressure	81.8	65.7
High Pressure	13.7	126.9

Q. DO THE COMPANIES SPECIFICALLY TARGET REMOVAL OF LPP WITHIN THE FLOODPLAIN?

A. No. As noted in the response to CNY-48, the Companies have "no distinct timelines for retirement of LPP by type or location." The Companies do, however, consider location within the FEMA's 2015 Preliminary Flood Insurance Rate Maps (PFIRMS) 1% annual chance floodplain (referred to hereafter as the "2015 FEMA 100-year floodplain") and the FEMA 2015 Preliminary Flood Insurance Rate Maps (PFIRMS) 0.2% annual chance floodplain (referred to hereafter as the "2015 FEMA 500-year floodplain") within their main replacement prioritization methodology. In particular, the prioritization methodology includes a 'tie-breaker' mechanism that gives priority to LPP in the 2015 FEMA 100-year or 2015 FEMA 500-year floodplain,[1] in the event that two or more sections of LPP have identical prioritization scores. According to KEDNY's response to CNY-14(i) and CNY-77, KEDNY will remove an additional 2.9 miles of low-pressure LPP in the 2015 FEMA floodplain under its Proactive LPP Program as a "direct result" of this 'tie-breaker' mechanism. KEDLI will remove 13.9 miles, according to its response to CNY-77, but it has not indicated whether this is the direct result of including this 'tie-breaker' mechanism. We understand that, each year, in addition to the LPP replaced pursuant to the Proactive LPP Program, both KEDNY and KEDLI will likely replace several miles of low pressure LPP in the floodplain

[1] Throughout this testimony, when we refer to the 2015 FEMA 100-year and 500-year floodplain, we are referring specifically to FEMA's 2015 PFIRMs 1% annual chance floodplain (100-year floodplain) and FEMA's 2015 PFIRMs 0.2% annual chance floodplain (500-year floodplain).

pursuant to other main replacement programs, including under the CSC Program. These efforts are evidenced in the Companies' response to CNY-76 where, for example, KEDNY indicated that it replaced nearly 24 miles of low pressure LPP in the 2015 FEMA floodplain in 2018, despite having no mechanism in place in 2018 to specifically target LPP in the floodplain.

Q. WHAT IS YOUR RECOMMENDATION?

A. The Companies should proactively improve their efforts to prioritize replacement of leak-prone services and mains in flood prone areas. Initially, the Companies should continue to utilize the tie-breaker mechanism in their main replacement prioritization methodology. However, the Companies' efforts should be bolstered such that each Company is required to achieve total replacement of low pressure LPP in the 2015 FEMA 500-year floodplain by 2030 (*i.e.*, within 10 years). This would require KEDNY and KEDLI to remove or replace approximately 8.2 miles and 6.6 miles per year, respectively, of low pressure LPP in the 2015 FEMA 500-year floodplain – an amount that should be readily achievable without any requirement for incremental funding. According to CNY-76, KEDNY has replaced an average of 10.7 miles per year of low pressure LPP in the 2015 FEMA floodplain[2] from 2014-2018, while KEDLI has replaced an average of 7.9 miles per year over the same timeframe.

[2] National Grid's discovery and testimony on this issue does not specify whether this is within the 2015 FEMA 100-year or 500-year floodplains, but based upon past conversations with National Grid, The City believes it is referring to the 2015 FEMA 500-year floodplain

Q. WOULD YOU SUPPORT AN ACCELERATED EFFORT TO REPLACE LOW PRESSURE LPP IN THE FLOODPLAIN?

Yes, we could support an incentive for accelerated low-pressure LPP replacement in the 2015 FEMA 500-year floodplain and the 2015 FEMA 100-year floodplain. Because of the added risk posed by low-pressure LPP in the floodplain, as discussed herein, an added incentive to hasten floodplain LPP removal could be justified. Thus, we propose a one BP PRA for each full incremental two miles of low-pressure LPP replaced in any given year, but only if the Companies exceed the minimum targets for low-pressure LPP in the floodplain recommended in this testimony, and only if that incremental low-pressure LPP is located within the 2015 FEMA 500-year floodplain (which also encompasses the 100-year floodplain). This incentive would be capped at five BPs. This proposal represents a more appropriate balance between ratepayer and shareholder interests.

B. STORM HARDENING

Q. DO YOU HAVE ANY RECOMMENDATIONS RELATED TO NATIONAL GRID'S STORM HARDENING PROGRAMS?

A. Yes. Many of our recommendations relate specifically to items that were addressed by the Companies in the "2017 Storm Hardening Collaborative Report," which was filed on April 26, 2018 in Cases 16-G-0058 and 16-G-0059. We specifically address four items that were discussed in the 2017 Storm Hardening Collaborative Report: (1) mini-gate and take stations located within the current 2015 FEMA 100-year and 2015 FEMA 500-year floodplains; (2) height of meter

sets in floodplains; (3) the flood study for the Greenpoint LNG plant site; and (4) remote shut-off valves. We also discuss our concerns with the Companies' proposed Expanded Residential Methane Detection program.

Q. WHAT ARE YOUR CONCERNS RELATED TO MINI-GATE AND TAKE STATIONS LOCATION WITHIN THE FUTURE 2050s FLOODPLAIN?

A. The 2017 Storm Hardening Collaborative Report identified 13 mini-gate and take stations that are located within the current 2015 FEMA 100-year and 500-year floodplains, as well as the future 2050s floodplain. Eleven out of these thirteen mini-gate and take stations are located within the current 2015 FEMA 100-year and 500-year floodplains. With regard to these 13 stations, National Grid agreed "[w]here possible and cost effective…[to] consider design changes to the height of telemetry cabinets using [design flood elevations] based upon the City's climate resilience design guidelines methodology for critical facilities with useful lives through 2050s."

Considering the asset class that is at stake – high- and medium-pressure mini-gate and take stations that are located within the current and/or future floodplains – it is extremely important that National Grid storm hardens these stations, including mitigating water intrusion into the underground vaults, elevating telemetry cabinets, and protecting vent poles to ensure that National Grid's control center retains the ability to remotely monitor and communicate with these assets during flooding events. Loss of communications during a flooding event leaves system operators blind to potential pressure build-ups and other issues

that could have the potential for severe neighborhood or system-level consequences. From the City's understanding of this list, we would request that National Grid in particular prioritize the eleven stations that are located in the current 2015 FEMA 100- and 500-year floodplains and mitigate these risks to these gate stations by 2025 at the latest, if not sooner.

Q. HOW IS NATIONAL GRID ADDRESSING THESE 13 MINI-GATE AND TAKE STATIONS?

A. Based upon the Companies' discovery responses, the City understands that one of the stations (Coney Island) from the list of eleven has been storm hardened, including elevating and/or protecting telemetry cabinets due to work done in 2018. Another one (Kings Plaza) is scheduled to be storm hardened in FY 23, and two others should be rehabilitated within a decade. Two new gate stations (Chelsea and Downtown Brooklyn) were constructed with storm hardening guidance in mind. National Grid claims that the other seven stations do not require elevation, and states that other regulator stations in the area can provide remote sensing data indicative of their functioning in the event of a weather event affecting these seven stations. National Grid provided this explanation in CNY-108.

Q. IS NATIONAL GRID PROPERLY ADDRESSING THESE 13 MINI-GATE AND TAKE STATIONS?

A. The City supports the work that National Grid has done to storm harden one existing gate station, as well as to apply storm hardening principles to its two new stations. However, we strongly recommend that the Companies accelerate the

timetable for the rehabilitation at Clifton Gate and Citizens Gate to be completed by 2025, rather than the next decade. With respect to the seven other stations out of the total eleven (that are in the current 2015 FEMA 100- and 500-year floodplains and were identified as vulnerable by the Storm Hardening Collaborative), the City notes that the Companies' response in City-108 is inconsistent with the information that was provided and analyzed during the Collaborative, in particular with respect to Design Flood Elevations of specific assets.

Accordingly, we request that National Grid, per the recommendations contained in its Storm Hardening Collaborative Report, prioritize storm hardening efforts at these stations, including elevation of the cabinets to protect the critical equipment contained in them by 2025. Alternatively, if the Companies have different data than what was shared with the City, showing that these stations are indeed at Design Flood Elevations that do not require additional storm hardening, then the City would request to review this data. Finally, while it is helpful that for some (but not all) of these stations, there are other regulator stations in the area that can provide remote pressure data indicative of the functionality of the station, this is not a blanket substitute for being able to communicate and monitor the performance of these critical assets, especially during a significant coastal storm event.

Q. WHAT ARE YOUR CONCERNS RELATED TO THE HEIGHT OF METER SETS IN THE FLOODPLAIN?

A. The City understands that National Grid installs approximately 500-1,000 new meter sets in the 2015 FEMA floodplains each year. Given the forecasted floodplain expansion in the coming decades, it is critical for National Grid to examine whether (and if so, how) the Company can mitigate flooding risk to these meter sets through common-sense actions like raising the standard height design where feasible. However, in response to CNY-14(j), National Grid reported that it is "not recommending increasing the standard height design for regulators and meter sets for new installations."

Q. WHAT DO YOU RECOMMEND?

A. National Grid should develop and adopt procedures that provide guidance for elevating new meter sets located in the floodplains, for both low-pressure and high-pressure services. We understand that a one-size-fits-all approach may not work due to factors such as different types of housing and building design, and location within the floodplain. We also agree and understand that there will be need for exceptions within any such guidance that accommodates building type, geography, etc. Nevertheless, the development and adoption of guidance procedures related to meter sets in the floodplain is an important step in addressing the risk of flooding. Moreover, as noted in Attachment 1 to CNY-14, adopting such procedures would be in line with the practice of other utility companies that serve customers located within floodplains.

Q. HAS NATIONAL GRID CONDUCTED A FLOOD STUDY FOR THE GREENPOINT LNG PLANT SITE?

A. Yes. Following the completion of the Storm Hardening Collaborative in 2018, KEDNY conducted a desktop flood study of the entire Greenpoint LNG Plant site. At the City's urging, this study was informed by the City's climate projections for future sea level rise, storm surge, and urban precipitation levels. KEDNY considered the City's modeled future 2050s floodplain analysis (a modeled floodplain by the New York City Panel on Climate Change that accounts for how the geographic scope of flooding will increase due to projected sea level rise) in addition to the City's Climate Resilience Design Guidelines,[3] and also considered a flooding scenario occurring during peak demand.

Q. DID THE 2018 GREENPOINT LNG PLANT SITE FLOOD STUDY ADDRESS THE CITY'S CONCERNS?

A. Not entirely, no. The Greenpoint LNG Plant sits along Newtown Creek in Brooklyn, in an area that, according to the City's projections, will become more susceptible to flooding events in the not-too-distant future. Eastern portions of the site are in the current 2015 FEMA 100-year floodplain, while southern and eastern portions of the site are in the current 2015 FEMA 500-year floodplain. However, given its proximity to water, the City's projections show that by both the 2020s and 2050s, significant additional areas of the Plant will be located within the 100-

[3] https://www1.nyc.gov/assets/orr/pdf/NYC_Climate_Resiliency_Design_Guidelines_v3-0.pdf.

year floodplain, with a significant incremental portion of the Plant site falling within the 2020s and 2050s 500-year floodplain as compared to today.[4]

As recognized by National Grid in its pre-filed direct testimony, the Greenpoint LNG Plant is a critical component of the Company's gas supply portfolio and operation network, and can provide a material portion of KEDNY's peak day demand. Loss of this plant during peak demand, perhaps due to a winter flooding event, could therefore have significant reliability and cost implications for National Grid's customers. The criticality of this asset is only magnified given the potential gas supply constraints facing National Grid's service territory. Unfortunately, recent winter storms have highlighted the all-too-real possibility of significant flooding events during the peak gas season. For example, in January and March of 2018, Boston Harbor experienced its first and third highest tides on record since records began nearly a century ago, resulting in inundation of local streets.[5]

Q. WHAT DO YOU RECOMMEND?

A. Given the criticality of the Greenpoint LNG Plant, especially in light of recent permitting decisions that will, at a minimum, delay the NESE pipeline, National

[4] *See* NYC Department of City Planning, NYC Flood Hazard Mapper, *available at* http://www1 nyc.gov/site/planning/data-maps/flood-hazard-mapper.page.

[5] Jeremy C. Flox and John Hilliard, Boston Globe, "Late Friday high tide was milder than the near-record tide from Friday morning" (March 3, 2018) ("Boston Harbor's tide registered at 14.67 feet shortly after 11:15 a m., the third-highest since record-keeping began in 1928, and just a few inches shy of the record of 15.16 feet, which the city set on Jan. 4 during its last major winter storm, according to the National Weather Service."), *available at* https://www.bostonglobe.com/metro/2018/03/03/late-friday-high-tide-was-milder-than-near-record-tide-from-friday-morning/ox46WrIHftityv294LvFwM/story html.

Grid should engage a professional third-party engineering consultant to conduct a more fulsome flood study of the Greenpoint LNG Plant site. This study should include, but not necessarily be limited to, the following components, which were not assessed in the initial National Grid study:

- Hydrological study (*e.g.*, how does the water flow and drain across the site in a flood event, in an intense precipitation event, and in a combined flood and intense precipitation event);
- Geotechnical condition (*e.g.*, assess soil composition to understand how the ground will hold/drain water, including potential contamination impacts to other areas/sites);
- What happens if multiple (not just one) assets are impacted;
- The combined risks at the site due to a flood event with intense precipitation;
- The potential risks of flood mitigation strategies upon drainage;
- The potential risks and physical impacts from wave action;
- The potential risks from high heat and high wind;
- Ground truth elevation data from the 2018 Greenpoint LNG Plant study;
- Ground truth/validate the mitigation strategies that are presented and present potential new recommendations based upon experiences and best practices elsewhere;
- Review potential unintended consequences or secondary effects to the proposed mitigation strategies; and

- Review power distribution across the site, including resiliency/reliability of substation L to support full existing load plus liquefier load, in the event substation M is out of service.

Again, given the criticality of the Greenpoint LNG Plant site, we recommend that this proposed study be completed as soon as possible, and in any event no later than the end of calendar year 2021.

C. ENHANCED RESIDENTIAL METHANE DETECTORS

Q. WHAT IS NATIONAL GRID'S PLAN REGARDING THE ENHANCED RESIDENTIAL METHANE DETECTION PROGRAM?

A. This Program would seek to install commercially available plug-in methane detectors in residential customer homes. It is an expansion of the Residential Methane Detection Pilot Program initiated in the 2016 Rate Cases and, beginning in the Rate Year and continuing for the next three years thereafter, would result in 54,000 detectors installed in KEDNY's service territory each year and 36,000 in KEDLI's each year, for a total planned deployment of 360,000 detectors. National Grid, on page 44 of its Gas Safety Panel testimony, proposes to specifically target "room set meter customers, low income customers, multi-use residences, and customers served by smaller diameter (two and four inch) cast iron piping." The detectors will be accompanied by an educational brochure and distributed in customer-facing interactions, such as service inspections.

Q. HOW DOES NATIONAL GRID PROPOSE TO IMPLEMENT THIS PROGRAM?

A. According to the information provided in the Gas Safety Panel's direct testimony and National Grid's response to CNY-128, the success of the Program is heavily dependent on customers installing and operating the devices, as well as understanding how the devices themselves work and the safety features provided. National Grid will distribute the detectors to customers along with a brochure and the manufacturer's instructions so that the customer may install it. National Grid proposes to send a follow-up letter and brochure, and each device will contain a unique bar code, which National Grid will scan into its system at the time of distribution, so that National Grid can track methane calls from households that have methane detectors. Otherwise, National Grid has proposed no other quality control or customer guidance linked to this program. Under this proposal, National Grid's job effectively ends once the detector is in the customer's hands and does not provide a centralized tracking mechanism for potential leaks.

Q. WHAT IS THE CITY'S ASSESSMENT OF NATIONAL GRID'S PROPOSAL?

A. Conceptually, we support this Program. Methane leaks are dangerous, and improving detection can protect customers, especially the most vulnerable that are being specifically targeted by this Program. Nevertheless, the Program's design and implementation, as proposed by the Companies, raises concerns.

Q. WHAT CONCERNS DOES THE CITY HAVE?

A. Essentially, there is no quality control, limited customer guidance, and no centralized tracking system. These issues could render an otherwise conceptually

helpful program ineffective because it is entirely dependent on the customer to properly install, maintain, and use the device, with little meaningful support from National Grid.

Q. WHAT QUALITY CONTROL ISSUES HAVE YOU IDENTIFIED?

A. National Grid does not know whether a detector works before delivering it to a customer. Considering that National Grid seeks to install 360,000 detectors, even a one percent error rate – not complete failure, but less-than-ideal performance – can place an unacceptable number of customers in jeopardy and reduce the ability of this Program to detect leaks. In addition, because the manufacturer's instructions direct residents to call 911 in the event that the alarm sounds, false positives due to device error or improper installation could place added unnecessary burdens onto the City's emergency response resources. Moreover, per the response to CNY-128, the detectors have a seven-year useful life, but the Companies have no plan to replace the detectors, meaning that the Program will end once the detectors no longer function.

Q. WHAT QUALITY CONTROL MEASURES WOULD HELP ADDRESS THESE CONCERNS?

A. A simple and effective measure would be to test a representative sample of detectors to ensure they work as intended, including the device's detection range and sensitivity to methane leaks not located on the customer's premises or methane from organic or biogenic sources. This also would include testing the end-of-life warning that the manufacturer's instructions indicate will sound after seven years,

per Attachment 3 to CNY-128. Seven years is a long time for a customer to remember that they must eventually replace their detector, and this delay can place customers in jeopardy if they assume that their detector works because no end-of-life alarm sounded.

Q. IS RELYING ON CUSTOMERS TO TEST THEIR OWN DETECTORS SUFFICIENT?

A. No. The brochure and manufacturer's instructions ask that the customer test their own detector weekly to make sure the alarm is sounding properly. However, it is too easy for a customer to forget to test their alarm every week, no matter how colorful the brochure, and it is reasonable to assume many recipients of these detectors will not test on a weekly basis. Considering that this is a ratepayer-funded program, ratepayers should have assurances that National Grid will make an attempt to ensure that the detectors at the heart of their Program work, even if customers forget to test them. Our recommendations in this regard are set forth below.

Q. COULD YOU ELABORATE ON THE LACK OF CUSTOMER GUIDANCE?

A. Yes. Per CNY-128, National Grid gives customers: (i) a brochure that summarizes what the detectors do and how to install them; (ii) the manufacturer's instructions for installation and all associated warnings; and (iii) the detector itself. Additionally, National Grid will follow up with a letter and a second brochure. Aside from this, there is apparently no other customer guidance.

The lack of customer guidance threatens to derail this Program. First, there is no assurance that the customer has actually installed a detector – the Company representative that provides the detector does not perform the installation, nor does that representative observe its installation. Second, no guidance is provided regarding correct installation and maintenance outside of the brochure and manufacturer's instructions. The 12-page set of instructions, provided as Attachment 3 to CNY-128, is itself a barrier. A customer is not likely to read such a document to ensure proper installation. Accordingly, the customer may miss important instructions such as: the manufacturer recommends that several detectors be installed; the detector cannot receive noisy power, such as if another device with a large load is on the same circuit; or that "turbulent air," such as from fans on a hot day, impede the functioning of the device. Customer testing of the device will not indicate to them whether it has been properly installed per this guidance. Moreover, customers have to fill out and place the labels that remind them what to do if the detector sounds. All of these instructions are buried in pages 3 through 5 of the manufacturer's instructions, and missing any one of these important points could render the detector ineffective.

Q. WHAT DO YOU RECOMMEND TO REMEDY THIS ISSUE?

A. We recommend two improvements. First, a Company representative should install the detector and provide concise verbal instructions regarding its use. Having already been tested, this would ensure that a functioning device is placed in the appropriate location. Additionally, it provides another means of informing the

customer of important information without requiring them to read a lengthy document.

Second, twice each year during the four-year deployment schedule, the Companies should test a sample of in-service detectors to ensure proper functioning. This will give the Companies an opportunity to identify and resolve any problems with the detectors, before wide-scale deployment is completed.

Q. WOULD THESE IMPROVEMENTS IMPOSE ADDITIONAL COSTS?

A. We do not believe so. Company installations and sample testing should be able to be accommodated within the budget requested.

Q. DO YOU HAVE ANY ADDITIONAL SUGGESTIONS RELATED TO THE RESIDENTIAL METHANE DETECTORS?

A. Yes. We understand that Con Edison is pursuing a similar, but different, program for residential methane detection, whereby the detectors will have the ability to communicate directly with Con Edison via the AMI meters and infrastructure that Con Edison is currently deploying. We understand that National Grid does not currently have any plans for deploying AMI meters. However, in the long term, there will be a need to monitor gas usage much more closely both from a safety and supply perspective. The Companies should therefore re-examine the potential deployment of AMI and, in doing so, examine whether they can utilize residential methane detectors that have the capability to sync up with AMI, to ensure consistency with Con Edison (and across the City).

D. REMOTE SERVICE SHUTOFF VALVES

Q. WHAT ARE THE COMPANIES' PROPOSALS REGARDING REMOTE SERVICE SHUTOFF VALVES?

A. On pages 85-87 of KEDNY's GIOP testimony and pages 78-80 of KEDLI's, the Companies propose to install remote service shutoff valves at customers located in the FEMA 100-year and 500-year floodplains. Per the response to DPS-561, the valves will be in both the 100-Year and 500-Year floodplain. These valves, National Grid avers, would use a communications network that would allow the valves, once they detect flooding, to selectively stop the flow of gas to areas affected by flooding without impacting other areas. In doing so, the Companies can prevent the negative effects of water infiltrating gas mains, which is discussed elsewhere in this testimony. This program was originally planned to begin in 2017, but lack of a commercially available shutoff valve delayed the program. KEDNY proposes to install 29,000 valves (an expansion from the original program proposal of 25,000 valves), and KEDLI 60,000.

Q. WHAT IS THE CITY'S ASSESSMENT OF THIS PROPOSAL?

A. Our response is twofold. First, we have concerns about how this program will be managed. Second, the data this program would generate presents an excellent opportunity for collaboration with the City. Despite the concerns about program management, given the potential benefits to facilitate faster service restoration and minimizing equipment damage, we recommend approving this proposal, while simultaneously granting the City the ability to leverage the data for its own flood-

monitoring programs. Moreover, we support the placement of valves in both the 100-Year and 500-Year floodplain.

Q. WHAT ARE THE CITY'S CONCERNS REGARDING PROGRAM MANAGEMENT?

A. Fundamentally, this is a nascent, untested technology, and the Program itself has already seen significant delays. Originally, as KEDNY and KEDLI discuss in their GIOP testimonies, they proposed to launch this Program in calendar year 2017. There was no commercially available shutoff mechanism at the time; the valves have only become available as of February 2019. The valves are truly brand new, too. Per CNY-78, the valves have only been tested at the manufacturer's facilities and not in a live situation. Indeed, the Companies have not yet completed their lab testing of the valves. Moreover, per CNY-78(d), no other gas utility is using these valves. Thus, this is a pilot not only for National Grid, but for the gas industry as a whole.

Considering past delays and the untested nature of the valves, it is reasonable to anticipate issues arising that could impact the effectiveness of this program. These include, but are not limited to, equipment failure in real world conditions and improper installations. These concerns are not theoretical. Several years ago, Con Edison installed a new technology, called a Vent Line Protector ("VLP") Device, which was designed to perform a similar function to the remote service shutoff valves; namely, protect gas customers during a flooding event. Over 80 percent (1,837 out of 2,204) of these VLP Devices were initially improperly

installed, and some of the devices were defective, thus requiring remedial action by Con Edison.[6] National Grid should take care to avoid a similar outcome with the remote sensor shutoff valves.

Q. SHOULD THIS PROPOSAL BE IMPLEMENTED DESPITE THESE ISSUES?

A. Yes. Despite our concerns, we agree that this program nevertheless has the potential to offer significant benefits, particularly during flooding events. Because of the storm hardening benefits and the opportunity for collaboration, which is described below, the City supports this proposal with certain enhancements. First, the Companies should implement a plan to methodically and robustly test these devices to ensure they operate as designed, and to gradually deploy such devices in phases. The Companies also need to perform outreach to educate customers on how these remote operated shut off valves will work, and what they are intended to prevent. The Companies should further confirm there will be no "accidental" automatic shutoffs of gas as a result of the operation of these sensors, and that they only will trigger if there is sufficient flooding to cause water intrusion and possible damage to customer equipment.

Q. HOW DOES THIS PROPOSAL PRESENT AN OPPORTUNITY TO SUPPORT THE CITY'S OWN FLOOD MONITORING EFFORTS?

[6] *See* Cases 13-E-0030, et al., Consolidated Edison Company of New York, Inc., Order Adopting Storm Hardening and Resiliency Collaborative Phase Three Report Subject to Modifications (issued January 25, 2016) at 26.

A. The valves activate when they detect flooding, and this information is relayed via a communications network, per the Companies' descriptions in their GIOP testimonies. The Companies propose to install 89,000 valves in total, which provides 89,000 data points on the location and timing of the floods, all in real-time. Furthermore, the Companies will only need to interrupt gas services to the specific areas affected, indicating that this data is quite granular, perhaps even to an individual sensor. This data can do more than just protect customers in the event of flooding – it can be used to track the extent, duration, and frequency of floods. This type of tracking is one of the primary objectives of the "Jamaica Bay Community Flood Watch Project" (the "Flood Watch Project").

Q. COULD YOU ELABORATE ON THE FLOOD WATCH PROJECT?

A. Yes. Due to climate change, the City is experiencing more regular and more extreme tidal flooding, called "sunny day flooding." As sea levels rise, this will only get worse, shifting the flood maps in unexpected ways. The Flood Watch Project seeks to document this sunny day flooding in the Jamaica Bay watershed through citizen science reports so that the City and its citizens can create maps and make predictions about the extent of flooding and its effects. The data the Remote Service Shutoff Valve Program will generate would be a valuable resource to help improve the Flood Watch Project's efforts to help improve forecasts of flooding and coastal hazards.

Q. WHAT DATA WOULD BE REQUIRED TO MAXIMIZE THE EFFECTIVENESS OF THE REMOTE SHUTOFF VALVE PROPOSAL?

A. The City would need access to the following data points for each valve in the City and the Rockaways:

- The latitude and longitude of each valve;
- The elevation of each valve; and
- The date, time, and duration of the flood event.

The City does not request this data in real time; rather, the City requests that National Grid share this data with the City in monthly reports, in a readily-accessible format like Microsoft Excel.

Q. WHY DO YOU SUPPORT VALVE PLACEMENT IN BOTH THE 100-YEAR AND 500-YEAR FLOODPLAINS?

A. Flooding is not just a problem in the 100-Year floodplain; flooding in the 500-Year floodplain has the same similar consequences despite being less frequent, and those customers therefore require the same protection. Importantly, as sea levels rise, there will be increasing likelihood of flooding in both the current 100-year and 500-year floodplains. Data from the valves will help the City better understand what areas are regularly experiencing more flooding due to climate change. Monitoring the growth of the floodplain is vital and can only be done if the valves are there to collect the data.

Q. COULD YOU SUMMARIZE THE CITY'S RECOMMENDATIONS ON THE PROPOSED REMOTE SERVICE SHUTOFF VALVE PROGRAM?

A. Yes.

- Because of the untested nature of the devices and National Grid's issues in getting the device to commercial viability in the past 3 years, the devices should be tested robustly and deployed gradually;
- Because of the protections it provides customers and the data it collects, the Program should go forward, provided that the devices are comprehensively tested, and customers are adequately educated about them;
- National Grid should share the information noted above with the City; and
- The devices should be installed in both the 2015 FEMA 100-year and 500-year floodplains.

CONCERNS ON IMPACTS TO NYC GREEN INFRASTRUCTURE ASSETS

Q. WHAT ARE NYC GREEN INFRASTRUCTURE ASSETS?

A. In New York City, Green Infrastructure, or GI, refers to an array of practices that use or mimic natural systems to manage stormwater runoff. Stormwater is either directed to engineered systems for infiltration or detained at a slower rate before it enters the combined sewer system. Types of Green Infrastructure include:

- Rain Gardens – vegetated or landscaped depressions in the sidewalk designed with an engineered soil that promote infiltration of stormwater runoff into the underlying soil.
- Stormwater Greenstreets – similar to Rain Gardens, but constructed in the roadway and usually larger than a Rain Garden.
- Green Roofs – roofs made up of a top vegetative layer that grows in an engineered soil, which sits on top of a drainage layer.

- Permeable Paving – using a range of materials and techniques, permeable pavers or porous concrete allows water to seep in between paving materials and be absorbed into the ground.

Through Green Infrastructure implementation, DEP is working toward a reduction of 1.67 billion gallons a year in Combined Sewer Overflows ("CSOs") by 2030. CSOs can negatively affect water quality and recreational uses in local waterways. Through the end of 2018, 4,585 Green Infrastructure Assets had been either constructed or were in construction. Approximately 5,500 Assets are planned to enter construction in 2019. A significant portion of these Assets are located in National Grid's service territory in Brooklyn and Queens.

Q. WHAT ARE YOUR CONCERNS RELATED TO GREEN INFRASTRUCTURE ASSETS AS THEY PERTAIN TO NATIONAL GRID?

A. Our primary concern is that NYC Green Infrastructure Program Assets are being damaged by National Grid street work. For example, in north Brooklyn, DEP has recorded damage to 47 Assets from National Grid work crews, totaling approximately $160,000 in damages. The damage is primarily from sediment and cement intrusion into the right-of-way Green Infrastructure Program Assets, which damages the engineered soils and plants located within the Assets. Other impacts include structural damage from National Grid pipe installations either through or adjacent to Green Infrastructure Program Assets. National Grid and its contractors should be following the green infrastructure protection stipulations

required by the DOT Street Opening Permit. It is clear to DEP that National Grid and its contractors are either unaware of these protection stipulations or have not fully adhered to them.

An additional concern about DEP Green Infrastructure Assets is limitations to siting and construction due to interferences with existing National Grid infrastructure. DEP will continue to review these locations with National Grid, but when there is no alternate location for the Asset, DEP retains the right to request National Grid to relocate their infrastructure. National Grid should respond within 10 business days to DEP's request for relocation. In the event that National Grid cannot move its infrastructure, National Grid must provide a specific in-depth justification. In instances where the infrastructure can be moved, it should be relocated in a mutually agreed upon timeframe.

For new National Grid infrastructure or relocation of existing infrastructure, National Grid should consider Green Infrastructure Program implementation, and the infrastructure should be installed with consideration to the future implementation of Green Infrastructure Assets. These considerations include placing natural gas lines six feet instead of three feet from the curb line when feasible, which will allow for DEP to install right-of-way Green Infrastructure Assets.

Q. ARE YOU REQUESTING MONETARY COMPENSATION FROM NATIONAL GRID AS PART OF THIS RATE CASE?

A. Our primary concern is to stop any future damage to Green Infrastructure Assets, and to ensure that all National Grid employees and contractors are properly informed, trained, and follow the protection stipulations required by the DOT Street Opening Permit stipulations. That being said, by submitting this testimony, DEP is not foregoing any right to seek damages for GI assets, and will continue to seek repairs in kind and restitution for damages to GI assets outside of this rate case.

Q. ARE THESE ASSETS VISIBLE TO NATIONAL GRID WORK CREWS?

A. Yes. The GI Program protection stipulations clearly describe the difference between a typical street tree and a DEP GI asset. DEP also provides a presentation on GI at the quarterly Department of Transportation (DOT) Construction Advisory Council Meetings. DEP is preparing additional training materials for National Grid and its contractors to eliminate any remaining confusion. In addition, DEP coordinates with National Grid extensively during the design process, and National Grid is aware of all locations that proceed to construction. Moreover, when GI Assets are in the vicinity of a DOT street opening permit site, the street opening permit includes a notification that indicates that GI Assets are nearby (see below):

BIOSW1	A NYCDEP RIGHT-OF-WAY GREEN INFRASTRUCTURE PRACTICE IS LOCATED ON THIS ROADWAY AND/OR SIDEWALK. PERMITTEE MUST PROTECT ALL NYCDEP GREEN INFRASTRUCTURE DOWNSTREAM OF THE WORK AND WITHIN 5 FEET OF THE WORK AREA. PERMITTEE MUST FOLLOW NYCDEP'S GREEN INFRASTRUCTURE PROTECTION REQUIREMENTS AT http://www.nyc.gov/html/dep/pdf/green_infrastructure/ogi-protection-procedure.pdf PERMITTEE IS RESPONSIBLE FOR RESTORATION OR RECONSTRUCTION OF DAMAGED DEP GREEN INFRASTRUCTURE AS DIRECTED BY DEP.

Finally, DEP maintains a web map that shows locations of built Green Infrastructure Assets, as well as those in construction, and this map can be utilized

by utility personnel to confirm Asset locations, and DEP will include directions to the web map in a forthcoming stipulation update. Despite these existing measures, however, DEP continues to experience issues with damage to Green Infrastructure Assets caused by National Grid work crews.

Q. WHAT DO YOU RECOMMEND TO REDUCE DAMAGE TO GREEN INFRASTRUCTURE ASSETS?

A. Our recommendations are three-fold. First, National Grid should notify DEP a minimum of two weeks prior to street openings, and DEP will send out an inspector during construction to ensure proper GI protections are implemented.

Second, National Grid should provide bi-annual training on identifying and protecting Green Infrastructure Assets to its employees and contractors using DEP-supplied training materials. These training materials are currently in development by DEP. DEP will also offer to provide 'train the trainer' training once per year to National Grid employees and contractors. National Grid should add this training as a requirement of the Companies' third party contracts.

Third, National Grid should identify a central point of contact for Green Infrastructure issues, including construction coordination and damage claims.

CONCERNS WITH PROPOSED GAS SAFETY METRICS AND PERFORMANCE MEASURES

Q. ARE THE COMPANIES SEEKING ANY PERFORMANCE INCENTIVES?

A. Yes. The Companies are proposing to modify the following performance measures: Leak Prone Pipe Replacement, Emergency Response, Leak Management, Damage Prevention, and Gas Regulations Performance.

Q. ARE THE PERFORMANCE INCENTIVES THAT THE COMPANIES SEEK JUSTIFIED?

A. No. Fundamentally, we disagree with the Companies' proposals, which essentially seek to reward them for doing basic prudent utility work. Several of the proposed performance metrics are coupled to shareholder incentives for certain natural gas distribution operations activities that should not be incentivized – emergency response and damage prevention – because these activities are within industry norms of utility operations as required by federal regulations.

The CFR provides minimum pipeline safety regulations governing, in part, the operations of a utility's natural gas distribution system (*See* 49 CFR Part 192). The CFR requires prompt and effective response to and reports of gas odor in or near a building, and to every type of emergency that involves or could involve natural gas or a natural gas pipeline facility. The effectiveness of an operator's emergency response program is a factor considered for risk reduction within the Distribution Integrity Management Program ("DIMP") regulations in 49 CFR Part 192 Subpart P. The DIMP regulations also require an operator to identify and implement measures to address risks to its system and ultimately to people and property. This includes performance of an effective leak management program, which provides that an operator act appropriately to mitigate the hazards

associated with each leak. For non-hazardous leaks, appropriate action may include scheduling the line for repair and continued reduction in leak count.

Other proposed performance metrics are tied to rewarding the Companies for accelerating the implementation of certain capital investment programs or modifying the damage prevention metrics to eliminate components that measure damages that are wholly within each Companies' control. Below is a description of our analysis of each proposed incentive, and an explanation of why such incentive should be rejected by the Commission.

For ease of reference, Exhibit__(NYCGISP-3) includes an overview of our recommendations for the Companies' various gas safety-related incentive mechanisms and safety metrics.

A. LEAK PRONE PIPE INCENTIVES

Q. WHAT ARE THE COMPANIES PROPOSING FOR THEIR LEAK PRONE PIPE RETIREMENT MILEAGE TARGETS AND INCENTIVES?

A. As discussed above, KEDNY is proposing to increase its minimum LPP replacement target by 5 miles per year, starting at its current minimum removal target of 60 miles and accelerating to 75 miles per year by CY 2022, with a cumulative minimum target of 310 miles for CYs 2020-2023. KEDLI proposes to maintain its current 150 miles per year minimum removal target, with a cumulative minimum target of 620 miles for CY2 2020-2023. The Companies propose to maintain the basic structure of this performance metric: each Company will be subject to an NRA should they fail to meet their minimum LPP mileage

target. The LPP mileage targets for the NRA are set well below the levels that will be funded in rates; for both Companies, the NRA is five miles below that which they request in rates. As noted earlier, the Companies also propose to continue the existing PRA of two BPs for each full mile of LPP removed above the level that is funded in rates, up to five miles (10 BPs).

The Companies propose to increase the NRAs for failure to meet their target from the current 8 BPs to a proposed 15 BPs, at the expense of the number of at-risk BPs for the Gas Regulations Performance Measure. The Companies are also proposing to continue its GSRS Mechanism to recover the carrying costs on incremental capital expenditures and O&M expenses associated with the replacement of main above the targets established.

Q. DO YOU OPPOSE THE COMPANIES' PROPOSALS?

A. Yes. First, increasing the NRAs for an incentive that the Companies have had no difficulty meeting at the expense of other NRAs is not appropriate and fails to serve the purpose of the NRA. As there is little likelihood that the Companies will incur their proposed 15 BP penalty, the reallocation does not "more fairly allocate the 150 at risk basis points," as the Companies suggest at page 48 of the Gas Safety Panel, but rather only reduces the likelihood that the Companies will be subject to potential NRAs. Furthermore, as discussed in more detail below, in light of the Order to Show Cause, it is not appropriate at this time to reduce the number of BPs at risk under the Gas Regulations Performance Measure.

Second, the Companies are already proposing increased levels of capital investment and O&M expense. The Companies have not set forth a compelling need to further increase cost burdens on customers by rewarding the Companies for replacing more mains. Moreover, the Companies have not justified that replacing additional mains is even something to be rewarded. Under the Companies' proposals, customers would bear all of the capital and O&M costs for the additional replacements, plus the shareholder incentive.

Q. WHAT DO YOU RECOMMEND?

A. We recommend that the NRAs at risk under the LPP Retirement performance measure not be modified and remain at 8 BPs. In addition, as discussed earlier in this testimony, the PRA for LPP main replacements should be rejected because the Companies have failed to make the case as to why ratepayers should bear such additional costs.

B. EMERGENCY RESPONSE TIME INCENTIVES

Q. WHAT CHANGES ARE THE COMPANIES PROPOSING TO THEIR EMERGENCY RESPONSE SAFETY PERFORMANCE MEASURE?

A. The Companies are proposing a series of positive incentives for their 30 minute emergency response rates. Additionally, the Companies propose to increase the number of at-risk BPs from 12 to 25 at the expense of the number of at-risk BPs for the Gas Regulations Performance Measure.

Q. DO YOU OPPOSE THE PROPOSALS?

A. Yes, they are not acceptable. The Companies should not be rewarded for simply maintaining a safe gas system, which is their fundamental role. Additionally, neither Company has received an NRA under this performance metric. Reallocating BPs to this measure is thus just as inappropriate here as it is for the LPP retirement metric – considering the Companies' past emergency response performance, the Companies' proposal then simply becomes a reduced penalty for violating the Commission's gas safety regulations.

Q. PLEASE PROVIDE THE BASIS FOR YOUR OPPOSITION TO THE COMPANIES' PROPOSED EMERGENCY RESPONSE INCENTIVE PROPOSALS.

A. As a threshold matter, it is the utility's responsibility to provide safe and reliable gas service, including responding to gas leaks in an appropriate time frame to minimize the harmful consequences of gas emergencies to its customers, its employees, the general public, property, and the environment. The Companies are proposing to revise the Emergency Response Safety Performance Measure so that the Company receives an incremental positive incentive for having met a 30 minute response rate to emergency calls.

The expectation to meet a response time target at or below 30 minutes is a current industrywide norm. Prudent operators will continue to strive to improve upon (lower) this norm. Federal regulations (49 CFR Part 192) require an operator to implement an emergency response program that will minimize the hazard

resulting from a gas pipeline (distribution system or transmission system) emergency by providing a prompt and effective response to gas emergencies.

Moreover, CFR §§ 192.605(a), 192.605(b), 192.605(c), and 192.615(a) require an operator to follow procedures for conducting operations, maintenance, and emergency response activities, including procedures for responding promptly to a report of gas odor inside or near a building. The emergency response procedures an operator is required to establish in writing and to follow include emergency plans requiring performance of prompt and effective response to a notice of every type of emergency that involves or could involve gas or a gas pipeline facility (*see* CFR §§ 192.605(c), 192.615(a)(3)).

The presumed expectation for the operator to minimize the risk from the activities of natural gas delivery is also set by the DIMP regulations (49 CFR § 192.1007). The specific contribution of prompt and effective emergency response to risk minimization is the reduction of the consequences of failures that result in gas release or to situations that could imminently result in gas release.

The regulatory requirements to provide a prompt and effective response to gas emergencies, and the inequity of rewarding shareholders for performance that the Companies are required by regulation to do, are the bases for our opposition to the Companies' proposed emergency response incentive proposal.

Q. DO YOU HAVE ANY ADDITIONAL CONCERNS REGARDING THE EMERGENCY RESPONSE TIME INCENTIVES?

A. Yes. The Companies propose a positive revenue adjustment of up to 6 BPs for increasing their 30 minute response rate: 2 BPs for responding to 86% to 88% of calls in 30 minutes, 4 for more than 88% to 90%, and 6 for more than 90%. Neither Company would have met this proposed incentive if it had been in effect during the current rate plan.

We note, however, that Con Edison is proposing an identical incentive mechanism in its ongoing rate case in Case 19-G-0066. According to Con Edison's annual Gas Safety Performance Measures reports for the past five years, Con Edison's actual results for a 30 minute response time have been far better than National Grid's – so much so that Con Edison would have received positive incentives without having to show any improvement, if the metrics proposed by National Grid had been in place in those past five years for Con Edison.[7] Indeed, Con Edison's lowest 30-minute response time in the past five years was in 2014, when it achieved an 87.8% response time. In contrast, KEDNY and KEDLI response times have consistently ranged in the high-70% to low-80% range.

Similar to several of the other gas safety and performance metrics discussed herein, we see no reason why National Grid's performance should lag behind that of Con Edison. Customers living in Brooklyn, Queens, and Staten Island should expect the same level of prompt response to emergencies that

[7] The City also disputed the emergency response time metric in the ongoing Con Edison gas rate case. *See* Case 19-G-0066, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Gas Service*, Direct Testimony of New York City Gas Infrastructure Panel (filed May 24, 2019), at pp. 17-19.

customers living in Manhattan and the Bronx have experienced in recent years. Indeed, given the lag in National Grid's recent performance as compared to Con Edison, the Commission should re-visit the minimum required response time for both KEDNY and KEDLI, so that it better aligns with what customers have come to expect in other boroughs.

Q. DO YOU HAVE A PROPOSAL FOR EMERGENCY RESPONSE TIMES FOR KEDNY AND KEDLI?

A. Yes. We propose the following required response times and associated NRAs:

Required Response Time ("RRT")	Calls that must be responded to within RRT	NRA for failure to meet RRT
30 minutes	80%	6 basis points
45 minutes	90%	4 basis points
60 minutes	95%	2 basis points

This proposal recommends a five percent tightening of the 30-minute response time, from 75% to 80%, while maintaining the 45-minute and 60-minute response times and percentages proposed by the Companies. This represents an improvement that should be achievable without the need for additional funding given each Company's recent historical performance. We note that this is still low compared to Con Edison's recent historical performance but nevertheless represents an incremental improvement.

We also recommend that the Commission reject the Companies' proposal to increase the BPs at risk under this performance measure and, instead, retain the total at-risk BPs at 12. Neither Company has had difficulty meeting these metrics in recent years. It would therefore be inappropriate at this time to shift at-risk BPs away from Gas Regulations Performance to Emergency Response Time Performance, especially given the recent Order to Show Cause discussed earlier.

Q. DO YOU HAVE ANY ADDITIONAL COMMENTS OR CONCERNS REGARDING EMERGENCY RESPONSE TIMES?

A. Yes. The City has concerns about how the Companies actually measures their emergency response times for the purpose of tracking this metric.

Q. HOW DO THE COMPANIES CURRENTLY MEASURE EMERGENCY RESPONSE TIME?

A. According to the response to CNY-126, the Companies track and measure emergency response times based on the time difference between when a leak call is first received, and the arrival on site of a "First Responder," that is, a National Grid employee "qualified for emergency response." This First Responder then transmits their arrival time on a mobile application that records an electronically time stamped arrival time.

Q. WHAT ARE YOUR CONCERNS WITH THE EXISTING METHODOLOGY?

A. It is not clear how the Companies designate an employee as a First Responder, or whether that First Responder has any authority to effect actual repairs to a gas leak

without calling for additional assistance to the scene, for example, supervisors or crews with the necessary authority, equipment, and resources to handle leaks.

Q. WHAT DO YOU RECOMMEND TO MITIGATE THIS ISSUE?

A. We recommend that the Commission examine the Companies' emergency response protocols in more detail to determine if changes are warranted, including whether response times should be measured from when a qualified response crew begins remediating a leak, as opposed to when a First Responder first arrives on the scene.

C. GAS LEAK BACKLOG INCENTIVES

Q. WHAT ARE THE COMPANIES' PROPOSALS FOR THEIR LEAK MANAGEMENT PERFORMANCE MEASURES?

A. The Companies have two related leak management performance measures: (1) NRAs for failure to meet a workable leak backlog target and a total leak backlog reduction target; and (2) PRAs for every 50 high-emitting type 3 leaks eliminated beyond their backlog targets. The Companies propose to maintain their workable leak backlog target and continue to reduce their total leaks at the same rate per year as in the current Rate Plan (150 per year for KEDNY and 750 per year for KEDLI). The Companies propose to increase the number of at-risk BPs for their NRAs from 12 to 15 by reducing the number of at risk BPs for their Gas Regulations Performance Measure. Finally, the Companies propose to maintain their PRAs and targets for high-emitting type 3 leaks.

Q. WHAT HAS BEEN NYCEM'S EXPERIENCE WITH LEAKING GAS MAINS?

A. We are aware of at least one gas explosion in New York City – at Lexington Avenue and East 54th Street – that has been attributed to the accumulation of gas in a subterranean structure from a gas main with a small leak. An investigation into that explosion revealed that the leak was classified as type 3 and therefore was not required to be repaired by Con Edison.

Q, WHAT HAPPENED IN THAT INCIDENT?

A. One City employee was seriously injured and two other City employees also were injured.

Q. WHAT ACTIONS, IF ANY, DID NYCEM TAKE AS A RESULT OF THAT INCIDENT?

A. NYCEM formed a working group comprised of City agencies, Con Edison, and National Grid to, among other things, examine the incidences of gas leaks, identify ways to improve communications among and between City agencies and the utilities regarding gas leaks, and recommend ways to enhance public safety.

Q. WAS NATIONAL GRID COOPERATIVE IN THIS EFFORT?

A. Yes. National Grid attended working group meetings, commented on the various versions of the report, and provided the working group with helpful data. We also understand that National Grid has been investigating ways to improve its ability to detect gas leaks during its leak surveys.

Q. WHAT ARE NYCEM'S CONCERNS WITH GAS ACCUMULATIONS IN SUBSURFACE STRUCTURES?

A. If there is a leak in a gas main, the emitted gas will travel to and collect in any voids. Many subsurface structures, such as manholes and service boxes, are not sealed. Gas can travel along pipes, mains, conduits, and ducts into these structures and accumulate. Over time, the potential exists for the gas to reach a concentration level at which it can become explosive in the presence of a flame or heat source. In the incident we discussed, a City crew was repairing potholes using a torch to soften the asphalt.

There are a number of electrical and plumbing activities in which a heat source is in or near one of these subsurface structures or sparks are produced from cutting metal. Based on these concerns, the City would like the Commission to take action to reduce, to the maximum extent possible, the potential for natural gas to accumulate in these structures and potentially cause explosions.

Q. DO YOU HAVE ANY PERSPECTIVE ON HOW OFTEN THESE CONCERNS ARISE WITHIN NEW YORK CITY OVER THE COURSE OF A YEAR?

A. When NYCEM investigated the Lexington Avenue incident, we learned that there were approximately 4,000 active gas leaks of all types and 423 leaks in subterranean structures as of January 2016. Both Con Edison and National Grid have taken action to reduce the number of gas leaks since 2016. Nevertheless, there is still significant room for improvement.

Focusing on National Grid, it reported to the Commission that as of the end of 2018, KEDNY had a total backlog of 2,803 gas leaks and KEDLI had a total backlog of 8,199 gas leaks, which is composed almost entirely of type 3 leaks, as opposed to workable (type 1, 2, or 2A) leaks. This is illustrated in the charts below:

KEDNY Historical Leak Backlog Performance[8]

Year	Workable Leak Backlog Target	Actual Workable Leak Backlog	Total Leak Backlog Target	Actual Total Leak Backlog
2017	≤ 25	8	≤ 3,550	3,118
2018	≤ 25	10	≤ 3,400	2,803

KEDLI Historical Leak Backlog Performance[9]

Year	Workable Leak Backlog Target	Actual Workable Leak Backlog	Total Leak Backlog Target	Actual Total Leak Backlog
2017	≤ 25	0	≤ 10,000	9,232
2018	≤ 25	0	≤ 9,250	8,199

[8] Cases 16-G-0058 *et al.*, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service*, Gas Safety Performance Metrics Report for Calendar Year 2017 (filed March 30, 2018) (hereinafter "2017 KEDNY Metrics Report"); *id.*, Gas Safety Performance Metrics Report for Calendar Year 2018 (filed March 29, 2019) (hereinafter "2018 KEDNY Metrics Report").z

[9] *Id.*, Gas Safety Performance Metrics Report for Calendar Year 2017 (filed March 30, 2018) (hereinafter "2017 KEDLI Metrics Report"); *id.*, Gas Safety Performance Metrics Report for Calendar Year 2018 (filed March 29, 2019) (hereinafter "2018 KEDLI Metrics Report").

Based on our review of the Companies' gas safety performance reports to the Commission for the past five years and DPS-114, which compiles this information, the leak backlog for each Company was in the thousands of gas leaks each year. Clearly, something needs to be done to change the status quo.

Q. WHAT DO YOU RECOMMEND?

A. We understand that as the Companies replace their most leak-prone pipe, the incidence of gas leaks should decline. However, reliance on the leak-prone pipe replacement program is not sufficient.

The City recommends that the Commission direct the Companies to reset their leak backlog targets to the actual leaks remaining at the end of CY 2018, and to then dramatically reduce their total leak backlog so that their backlogs are negligible within five years. Using actual leak backlog numbers from the end of CY 2018 as the starting point, this would require a reduction in the backlog of about 560 per year for KEDNY and about 1,640 per year for KEDLI.

We also recommend that the Companies continue to reduce their backlog of workable leaks (those that could become hazardous over time) to zero, and then move to repairing all potentially hazardous leaks when they occur. Accordingly, the workable leak backlog target should be set at less than or equal to 10 per calendar year, as opposed to the 25 proposed by the Companies.

Because we are proposing to significantly increase the overall reduction in the leak backlog, the Companies' proposed increase in at-risk BPs for the leak backlog metric is appropriate as it helps motivate the Companies to achieve these

enhanced targets. At the same time, because the Total Leak backlog is so much higher for National Grid than for Con Edison, and because National Grid has successfully mitigated its Workable Leak backlog in the last couple of years, we propose to switch the BPs at risk for the Total Leak and Workable Leak metrics, such that 10 BPs are at risk for failure to meet the Total Leak metric (versus 5 BPs as proposed by the Companies) and 5 BPs are at risk for failure to meet the Workable Leak metric (versus 10 BPs as proposed by the Companies). If our proposal to modify the targets is not accepted, the increase in at-risk BPs should not be approved.

Q. DO YOU BELIEVE THAT IT IS POSSIBLE FOR NATIONAL GRID TO ACHIEVE YOUR PROPOSED LEAK BACKLOG THRESHOLDS USING PROPOSED FUNDING AMOUNTS SET FORTH IN THE COMPANIES' PRE-FILED DIRECT TESTIMONY?

A. Yes. Both Companies have made significant reductions in their total leak backlogs since 2014 – KEDNY has eliminated on average 316 leaks per year, and KEDLI has eliminated on average of 830, with both showing greater reductions in recent years as Proactive LPP replacement programs continue to ramp up. Combined with the synergies from replacing LPP, we believe that both Companies are capable of meeting these targets (particularly if, as proposed, KEDNY's Proactive LPP replacement program continues to ramp up over the next several years).

Moreover, we are proposing to reset the leak backlog targets to the actual leaks remaining at the end of CY 2018 (2,803 for KEDNY; 8,199 for KEDLI). For

KEDNY, at least, the Company should already be well below this target by the time the Rate Year starts. KEDNY reported in response to CNY-38 that its total leak backlog as of June 11, 2019 was 2,467, which is 336 leaks below our proposed reset of 2,803. KEDNY has another seven months before the Rate Year commences to lower its actual backlog even further, which should make these proposed annual reductions that much easier to attain.

We understand that the Company identifies and repairs hundreds of leaks of all types each year. However, it has been doing so for many years, and it still has been able to eliminate a significant portion of its leak backlog as well. It may take five years to fully eliminate the backlog under our proposal, but that elimination should be the goal. Put another way, the goal should not be to simply maintain thousands of minor leaks each year (i.e., the status quo).

To further incentivize eliminating gas leaks, we would also support a PRA that incentivizes the Companies to eliminate the backlog even faster than our proposed timeline. We suggest the following incentive mechanism: one BP for every 10 percent above our proposed yearly leak reduction, capped at five BPs. For example, if KEDNY eliminated 616 leaks instead of 560, it would receive 1 BP; if KEDLI eliminated 2,460 instead of 1,640, it would receive five BPs.

Finally, because we are proposing to materially shorten (and ultimately eliminate) the overall leak backlog, we see no reason to continue the existing PRA for Type 3 'high emitters.' Under our proposal, all leaks will be addressed in a prompt manner, and thus there is no reason to single out a specific category of leak

for special treatment. This has the added advantage of addressing gas leaks in subterranean structures, explained above.

Q. PLEASE DISCUSS THE BENEFITS TO THE PUBLIC OF ELIMINATING GAS LEAKS.

A. There are two primary benefits to doing so. The first is public safety. If gas leaks are fixed as they are identified, there would be far less opportunity for gas to migrate to subsurface structures and accumulate to explosive concentrations. Although gas explosions are rare in New York City, every recent gas explosion has resulted in injuries or death for one or more people. These injuries and deaths are avoidable, and reasonable efforts should be taken to avoid them. The death of a single New York City resident was sufficient cause for the Commission to institute rigorous stray voltage inspection and repair requirements. The Commission should take a similarly aggressive approach with respect to eliminating gas leaks.

Q. WHAT IS THE SECOND PRIMARY BENEFIT?

A. Methane is a very potent greenhouse gas, many times more potent than carbon dioxide. Eliminating gas leaks will reduce fugitive emissions of natural gas into the atmosphere and help to combat global warming. It is our understanding that the State, and the Commission in particular, are taking multiple actions to reduce greenhouse gas emissions, including increasing reliance on renewable and carbon-free resources and seeking substantial expansion of energy efficiency measures. As part of its portfolio of actions, the Commission can, and should, direct National

Grid and the other gas utilities to take steps to reduce and eliminate natural gas emissions from pipe leaks to the maximum extent practicable.

Q. ARE YOU AWARE OF THE PRACTICES OF OTHER UTILITIES REGARDING THEIR GAS LEAK BACKLOG?

A. We are not aware of the details of all utilities, but within New York State we are aware that the practices vary. For example, on May 20, 2019, New York State Electric & Gas Corporation ("NYSEG") and Rochester Gas and Electric Corporation ("RG&E") filed gas rate cases with the Commission in Cases 19-G-0379 and 19-G-0381, respectively. Their filings included pre-filed direct testimony from their Gas Operations Panels. Both Panels testified that the utilities have corporate requirements to repair all gas leaks of all types each year and strive to satisfy these requirements. According to those Panels' exhibits, their leak backlogs are minuscule: across all types of leaks (1, 2A, 2, and 3), NYSEG and RG&E have a three-year average of 10 leaks and 12 leaks, respectively. We understand that the NYSEG and RG&E leak backlog figures may be influenced by the character of their service territories, which may differ from the urban setting that comprises the KEDNY and KEDLI service territories, but the wide differences between NYSEG/RG&E's backlogs and National Grid's are significant enough to warrant mentioning.

We note further that on March 1, 2019, Con Edison reported to the Commission in its report on Gas Safety Performance Measures for 2018, filed in Case 16-G-0061, that as of the end of 2018, it had a total backlog of 283 gas leaks.

This is nearly 10 times lower than KEDNY's leak backlog as of June, 2019. In prepared direct testimony submitted on May 24, 2019 in Case 19-G-0066, the City is similarly recommending that Con Edison reduce its leak backlog to zero.

Q. DO YOU HAVE ANY CONCERNS ABOUT THE MANNER IN WHICH NATIONAL GRID CONDUCTS LEAK SURVEYS?

A. Yes. Following the Lexington Avenue gas explosion discussed above, NYCEM assembled a Natural Gas Interagency Working Group ("Working Group") that was charged with, among other things, improving interagency and utility communication related to active gas leaks that may impact NYC personnel and the general public. As part of that effort, the Working Group identified significant differences in the manner in which NYC's two gas utility companies – Con Edison and National Grid – perform gas leak surveys. Based on Companies' response to CNY-120, National Grid has made improvements in its gas leak survey activities to more closely align with Con Edison's practices. However, significant differences remain between the practices of National Grid and Con Edison, including:

- Con Edison performs monthly mobile 'low speed' surveys of gas mains, which KEDNY performs annually and which KEDLI does not perform at all; and
- Con Edison and KEDLI perform walking surveys of gas service connections in non-business districts every three years, while KEDNY performs these surveys every five years.

Q. WHAT IS YOUR RECOMMENDATION?

A. National Grid's leak survey practices should be aligned both internally and with Con Edison, so that the entire City is assessed in a consistent fashion. For example, KEDNY should conduct monthly mobile 'low speed' leak surveys, and KEDLI should perform similar surveys in the Rockaways portion of its service territory. In addition, KEDNY should perform walking surveys of gas service connections in non-business districts every three years, to align its practices with Con Edison and KEDLI.

D. DAMAGE PREVENTION SAFETY METRICS

Q. WHAT ARE THE COMPANIES' PROPOSALS FOR THEIR DAMAGE PREVENTION SAFETY PERFORMANCE MEASURES?

A. The Companies propose to: (1) increase the total number of at-risk BPs for this measure from 18 to 20 at the expense of the NRAs under the Gas Regulations Performance Measure; (2) remove the distinction between types of damages by only measuring overall damages per 1,000 one-call tickets, which would include third-party damages and no calls in addition to mismarks and company and contractor damages; and (3) add a PRA of up to 6 BPs for reducing overall damages below a certain threshold.

Q. DO YOU OPPOSE THE COMPANY'S PROPOSAL?

A. Yes. We oppose all three changes.

Q. WHY DO YOU OPPOSE THE REALLOCATION OF BASIS POINTS?

A. Increasing the at-risk BPs again is inappropriate – this is a reduction in the penalties for the Gas Regulations Performance Measure, nothing more. We can support increasing the BPs at risk for this performance measure, but not at the expense of other NRAs.

Q. WHY DO YOU OPPOSE THE CONSOLIDATION OF THE METRICS INTO JUST OVERALL DAMAGES?

A. Lumping all damages into overall damages is inappropriate. It eliminates the important purpose served by measuring any damage caused by the Companies not properly labelling their systems (mismarks), or caused by the Companies damaging their own system (company and contractor). These damages are wholly within the Companies' control and represent competent management of their systems to which all ratepayers are entitled. Obfuscating how well the Companies manage their systems would serve no purpose, and overall damages are already reported, so this proposal does not represent an improvement on the performance measures already in place.

Q. DO YOU HAVE CONCERNS ABOUT HOW THE COMPANIES COMPUTED THEIR RECENT PERFORMANCE ON THIS METRIC?

A. Yes. Both Companies may have misreported their 2018 performance on the damage prevention metric. According to the *2018 Pipeline Safety Performance Measures Report* ("Safety Report"), issued June 13, 2019, in Case 19-G-0298, KEDNY and KEDLI were required to omit retransmits or refreshes from their performance calculations since Staff issued its guidance in December 2015, as

noted on pages 9-10 of the Safety Report. However, the Companies did not do so, making year-to-year comparisons difficult, as Staff describes on pages 4 and 5 of the Safety Report. The inclusion of retransmits and refreshes artificially deflated the actual damages per 1,000 tickets. The table below, which compares the Companies' reported performance in each of their 2018 Metrics Reports and Staff's findings in the Safety Report for 2018, exemplifies this discrepancy:

KEDNY and KEDLI 2018 Damage Performance – Company Report vs. Staff Report vs. Targets

Metric (per 1,000 tickets)	KEDNY			KEDLI		
	KEDNY Report	Staff Safety Report	2018 Performance Target	KEDLI Report	Staff Safety Report	2018 Performance Target
Overall Damages	1.17	2.56	≤ 1.87	1.70	2.79	≤ 2.48
Mismark Damages	0.34	0.75	≤ 0.42	0.49	0.79	≤ 0.6
Company and Contractor Damages	0.01	0.03	≤ 0.09	0.09	0.15	≤ 0.13

This discrepancy provides independent grounds for opposing the proposed metric consolidation. The questionable quality of historical information the Companies have provided makes future reporting of all metrics even more important.

Q. SHOULD THE COMPANIES RECEIVE A PRA FOR THIS METRIC?

A. No. The Companies should not be rewarded for minimizing damage to their systems, especially for correctly marking the location of and not damaging their own facilities. This is part of competent system management.

Q. DO THE DISCREPANCIES BETWEEN THE COMPANIES' METRICS REPORTS AND STAFF'S SAFETY REPORT MAKE A REQUEST FOR A PRA INAPPROPRIATE?

A. Yes. The Companies' inclusion of retransmits artificially deflated their numbers, making them appear much lower than what Staff considers them to be. Whatever the case, this discrepancy prevents accurate evaluation of the Companies' historical performance. Until we can accurately measure the Companies' performance relative to their historical performance, there is no basis for shareholder bonuses as there is no way to be assured that there is exceptional performance to reward. Additionally, the inclusion of retransmits calls into question the ability of the Companies to properly compute their metrics. Until there is a consistent record of proper computation, there is again no basis to determine whether shareholder incentives are warranted.

Q. WHAT DO YOU RECOMMEND?

A. Our response is threefold. First, the number of at-risk BPs should not be increased if it reduces those at risk to the Gas Regulations Performance Measure. Our proposal also is to keep 4 BPs at risk for Overall Damages, 10 BPs at risk for Mismarks, and 4 BPs at risk for Company and Contractor.

Second, the current Rate Plan's damage performance categories – total damages, mismarks, and company and contractor – should be maintained, and the thresholds should continue to decrease to encourage better performance. Customers expect continual improvement, so we propose to continue the trend from the current Rate Plan when setting performance metrics. Our proposal can be found in Exhibit__(NYCGISP-3). These targets are based on excluding resent, refreshed, revised, retransmitted, reissued, or relocated one-call tickets, per the Commission's directive in its *Order Approving Implementation Plans*, issued March 10, 2017 in case 13-M-0314, at page 4, footnote 3.

Third, damage prevention, which represents competent management of the Companies' systems, should not be rewarded with PRAs. Considering the Companies' deflation of their damages, there needs to be sufficient time to ensure that the Companies are properly computing their performance and that there is enough data to properly evaluate a PRA. We consider three years or the next rate plan, whichever is later, an appropriate time to develop accurate data that may form the basis for a future request for a PRA.

Q. DO YOU HAVE ANY OTHER CONCERNS?

A. Yes. In addition to raising doubts about KEDNY's and KEDLI's actual performance, the Safety Report suggests that KEDNY and KEDLI performed much worse than they indicated in their 2018 Metrics Reports (and possibly older Metrics Reports). This is illustrated in the table above. The Companies' performance was subject to an NRA, and according to the numbers in the Safety Report, KEDNY potentially failed on all but the company and contractor damages sub-metric in 2018, and KEDLI potentially failed on all three, resulting in a possible NRA of 14 BPs for KEDNY and 18 BPs for KEDLI.[10] The Companies should provide a full accounting of whether they properly computed their performance and whether they should be subject to an NRA.

E. GAS REGULATIONS PERFORMANCE METRICS

Q. WHAT DO THE COMPANIES PROPOSE REGARDING THEIR GAS REGULATIONS PERFORMANCE MEASURES?

A. The Companies propose the following changes to their Gas Regulations Performance Measures: (1) decreasing the total number of BPs at risk from 100 to 75; (2) creating broader "tiers" so that repeated violations of Commission regulations would entail a smaller NRA; and (3) further dividing the types of violations into categories based on whether they are found through Records Audits or Field Audits.

Q. HOW DO THE COMPANIES JUSTIFY THEIR PROPOSED CHANGES?

[10] *See* Case 16-G-0058 *et al.*, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service*, Joint Proposal (filed September 7, 2016), at page 52; *See id.* at page 100.

A. On page 62 of their Gas Safety Panel, the Companies claim that, because they have improved since 2015, the above changes are appropriate, and on page 48, they describe the proposed reallocation of the BPs at risk as a more fair allocation.

Q. DO YOU FIND THESE CHANGES AND THE COMPANIES' JUSTIFICATION FOR THE CHANGES REASONABLE?

A. No. The Order to Show Cause, which details 1,616 alleged safety-related violations so far, not only calls into question National Grid's ability to safely and prudently manage its system and capital budget, but also the justification for the changes to the Companies' Gas Regulations Performance Measures. The Order to Show Cause highlights a multitude of serious, credible alleged violations of the very same regulations that are the subject of this performance measure. If there has been an improvement since 2015, as National Grid claims, it has not been to a level that could justify reducing the incentive for National Grid to comply with the Commission's safety regulations.

Furthermore, the changes appear to be less about fairness and more about shielding the Companies from the ramifications of their actions. For example, on pages 19-20 of the Order to Show Cause, the Commission notes 499 alleged violations of 16 NYCRR § 255.604, which is one of the "high risk" violations in the current measure, according to page 1 of Appendix 9 to the Joint Proposal in the current rate plan. These allegations, if proven and not subject to a separate penalty proceeding, could impose a significant NRA on National Grid. Combined, if all 1,616 allegations are confirmed, the Companies could be subject to their

entire NRA of 100 BPs. Under the Companies' proposal, going forward the penalty could only be 75 BPs, and perhaps even less. National Grid's changes therefore amount to nothing more than reducing its exposure to penalties.

Accordingly, National Grid's proposed changes to the Gas Regulations Performance Metrics lack justification. The purpose of this performance measure is to incentivize KEDNY and KEDLI to maintain a safe and reliable gas system, and this NRA must remain in place to do so. Indeed, changes to increase National Grid's potential penalty exposure under this measure could be justified in light of the allegations in the Order to Show Cause. For example, National Grid acknowledges on page 62 of its Gas Safety Panel testimony that, as a result of the 2016 KEDNY and KEDLI Rate Cases, there is a cap of ten occurrences per year for non-compliance with each regulation, which it claims helps to alleviate the "overly punitive" nature of the prior metric that allowed for unlimited occurrences. In light of the allegations in the Order to Show Cause – again, which may grow as Staff discovers more – the cap may not deliver the proper incentive.

Our only caveat to our position is that we would support shifting 3 at-risk BPs from the Gas Regulations Performance Measure to the Leak Backlog NRA, provided that our recommendations are adopted to reduce the Workable Leak and Total Leak targets.

RETURN THE HISTORIC NRA BALANCE

Q. WHAT ARE THE COMPANIES' REGULATORY LIABILITIES FOR NEGATIVE REVENUE ADJUSTMENTS IN PRIOR PERIODS?

A. According to the response to CNY-1, KEDNY is carrying a regulatory liability of $21.2 million based on NRAs that were incurred in 2011 and 2013-2017, while KEDLI is carrying a regulatory liability of $13.9 million based on NRAs that were incurred in 2011-2014. As of the date of this testimony, data was not available to determine whether any NRAs have accumulated based on 2018 performance; however, the Order to Show Cause suggests that penalties may be likely.

The Companies propose, at most, to use a small portion of their historical NRA balance to fund two proposed programs: (i) the Residential Methane Detector Program; and (ii) the incremental I&R Control Line Survey and GPS Mapping Program. According to KEDNY Exhibit___(GIOP-8CU) and KEDLI Exhibit___(GIOP-7CU), the total cost for these two programs over the Rate Year and Data Years is $9.08 million and $5.56 million, respectively. Thus, if the Companies' proposals were adopted, KEDNY and KEDLI would retain historic NRA balances of $12.02 million and $8.34 million.

Q. ARE THE COMPANIES' PROPOSALS FOR THE USE OF THESE NRAs APPROPRIATE?

A. No. There is no reason that either Company should be retaining money owed to customers for penalties that were incurred up to eight years ago. Moreover, use of the NRAs should not be conditioned upon, or tied to, specific capital programs.

Q. WHAT ARE YOUR RECOMMENDATIONS?

National Grid should immediately return the full balance of historic NRAs, with interest, to customers. These amounts should be applied as a credit against the

overall revenue requirement so that they are not dependent on use for any specific capital program.

Q. DOES THIS CONCLUDE YOUR TESTIMONY?

A. Yes.